



ASSIGNMENT No: 7

TITLE: Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Code

```
interface VehicleCar{
    void start();
}

class Car implements VehicleCar{

    public void start() {
        System.out.println("Car is starting.....");
    }
}

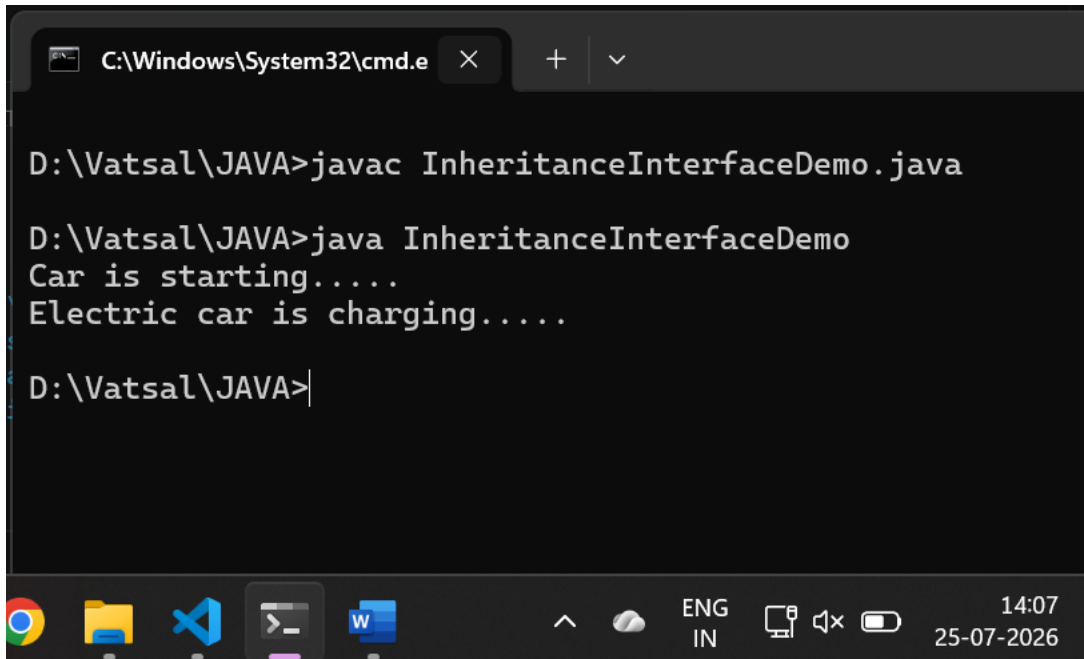
class ElectricCar extends Car{

    void charge() {
        System.out.println("Electric car is charging.....");
    }
}

public class InheritanceInterfaceDemo {
    public static void main(String[] args) {
```

```
ElectricCar e = new ElectricCar();
e.start();
e.charge();
}
}
```

Output

A screenshot of a Windows command prompt window. The title bar shows the path 'C:\Windows\System32\cmd.e'. The command prompt is open at the directory 'D:\Vatsal\JAVA'. The user has entered the command 'javac InheritanceInterfaceDemo.java' and then 'java InheritanceInterfaceDemo'. The output of the program is displayed: 'Car is starting.....' followed by 'Electric car is charging.....'. The command prompt is now ready for the next command.

```
C:\Windows\System32\cmd.e
D:\Vatsal\JAVA>javac InheritanceInterfaceDemo.java
D:\Vatsal\JAVA>java InheritanceInterfaceDemo
Car is starting.....
Electric car is charging.....
D:\Vatsal\JAVA>
```

Exercise

1. Create a **Shape application** where Circle and Rectangle inherit from Shape and calculate area.

Code

```
interface Shape {
    void calculateArea();
    void displayShape();
}

class Circle implements Shape {
    double radius;

    Circle(double radius) {
        this.radius = radius;
    }
}
```

```

    public void displayShape() {
        System.out.println("\nDisplaying Circle with radius: " + radius);
    }

    public void calculateArea() {
        double area = Math.PI * radius * radius;
        System.out.println("Circle Area: " + area);
    }
}

class Rectangle implements Shape {
    double width;
    double height;

    Rectangle(double width, double height) {
        this.width = width;
        this.height = height;
    }

    public void displayShape() {
        System.out.println("\nDisplaying Rectangle with width: " + width + " and height: " + height);
    }

    public void calculateArea() {
        double area = width * height;
        System.out.println("Rectangle Area: " + area);
    }
}

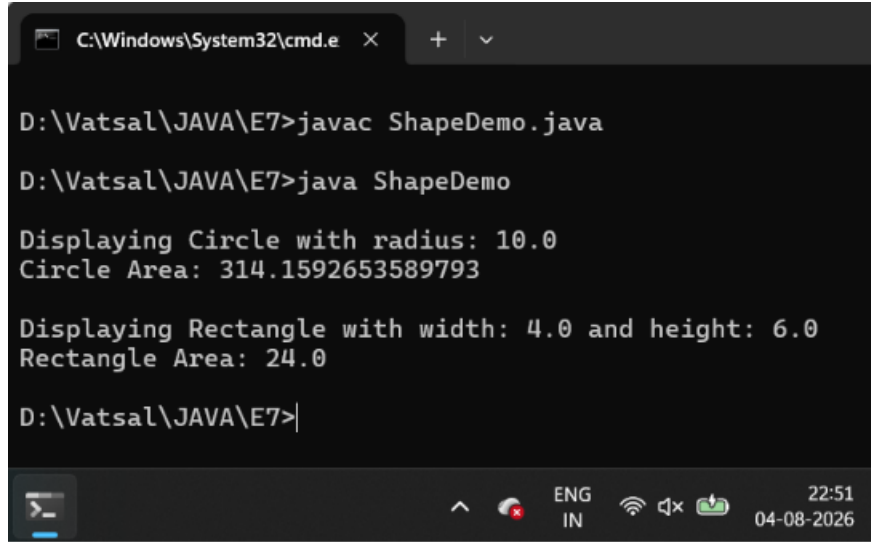
// Main Class
public class ShapeDemo {
    public static void main(String[] args) {

        Shape circle = new Circle(10.0);
        circle.displayShape();
        circle.calculateArea();

        Shape rectangle = new Rectangle(4.0, 6.0);
        rectangle.displayShape();
        rectangle.calculateArea();
    }
}

```

Output



```
C:\Windows\System32\cmd.e X + v

D:\Vatsal\JAVA\E7>javac ShapeDemo.java

D:\Vatsal\JAVA\E7>java ShapeDemo

Displaying Circle with radius: 10.0
Circle Area: 314.1592653589793

Displaying Rectangle with width: 4.0 and height: 6.0
Rectangle Area: 24.0

D:\Vatsal\JAVA\E7>|

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```

2. Develop a **E-Commerce Product System** where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

Code

```
interface Product {
    void displayDetails();
}

class BaseProduct implements Product {
    int id;
    String name;
    double price;

    public void displayDetails() {
    }
}

class ElectronicProduct extends BaseProduct {

    ElectronicProduct(int pId, String pName, double pPrice) {
        id = pId;
        name = pName;
        price = pPrice;
    }

    public void displayDetails() {
        System.out.println("\n--- Electronic Product ---");
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Price: " + price);
    }
}
```

```

}

class ClothingProduct extends BaseProduct {

    ClothingProduct(int pId, String pName, double pPrice) {
        id = pId;
        name = pName;
        price = pPrice;
    }

    public void displayDetails() {
        System.out.println("\n--- Clothing Product ---");
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Price: " + price);
    }
}

class GroceryProduct extends BaseProduct {

    GroceryProduct(int pId, String pName, double pPrice) {
        id = pId;
        name = pName;
        price = pPrice;
    }

    public void displayDetails() {
        System.out.println("\n--- Grocery Product ---");
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Price: " + price);
    }
}

public class ECommerceDemo {
    public static void main(String[] args) {
        ElectronicProduct laptop = new ElectronicProduct(101, "Laptop", 80000.0);
        ClothingProduct tshirt = new ClothingProduct(105, "Shirt", 750.0);
        GroceryProduct milk = new GroceryProduct(108, "Milk", 80.0);

        laptop.displayDetails();
        tshirt.displayDetails();
        milk.displayDetails();
    }
}

```

Output

```
C:\Windows\System32\cmd.e X + v

D:\Vatsal\JAVA\E7>javac ECommerceDemo.java

D:\Vatsal\JAVA\E7>java ECommerceDemo

--- Electronic Product ---
ID: 101
Name: Laptop
Price: 80000.0

--- Clothing Product ---
ID: 105
Name: Shirt
Price: 750.0

--- Grocery Product ---
ID: 108
Name: Milk
Price: 80.0

D:\Vatsal\JAVA\E7>
```

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